Constraints

$1 \leq N \leq 10^6$

$1 \leq a[i] \leq 10^9 \ \forall \ i \in [1, N]$

Sample input

```
3
2 1000000 999999
```

Sample output

```
2
```

Connected components

Given an undirected graph consisting of *N* nodes. There is a number written on each node which is given in the form of an array *a* of *N distinct integers*, where *a[i]* represents the number written on the i^{th} node.

There is an edge between the node i and j if $lcm(a[i], a[j]) \leq 10^6$.

Where LCM of two numbers is the smallest positive integer which is divisible by both of them.

Print the number of connected components in this graph.

Notes

- *1-based* indexing is followed.
- A connected component of an undirected graph is an induced subgraph in which any two vertices are connected to each other by paths, and which is connected to no additional vertices in the rest of the graph.

Function description

Complete the function *solve*. This function takes the following *2* parameters and returns the required answer:

- *N*: Represents the size of array *arr*
- *arr*: Represents the elements of array *arr*

Input format for custom testing

Note: Use this input format if you are testing against custom input or writing code in a language where we don't provide boilerplate cod

- The first line contains a single integer *N*.
- The second line contains *N* space-separated integers i^{th} of which is *a[i]*.

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- The second line contains *N* space-separated integers i^{th} of which is *a[i]*.

Explanation

Given

- N = 3
- a = [2, 1000000, 999999]

Approach

- For i=1 and j=2, lcm(2, 1000000) <= 1e6. Therefore there is an edge between 1 and 2.
- For i=2 and j=3, lcm(1000000, 999999) > 1e6. Therefore no edge between 2 and 3.
- For i=1 and j=3, lcm(2, 999999) >1e6. Therefore no edge between 1 and 3.
- The components are [1, 2] and [3]. Hence the answer is 2.

The coin problem

Given *N* coins whose amount ranges from *0* to *N-1* respectively. Your friend wants to take *K* coins out of your coins. You can only give the coins if the set of *K* coins is useful.

A set of coins is useful if the sum of the coins is divisible by a given integer *M*.

Find the number of ways in which your friend can get *K* coins. Since the answer can be large, print the answer modulo $10^9 + 7$.

Function description

Complete the *solve* function provided in the editor. This function takes the following *3* parameters and returns the answer:

- *n*: Represents the number of coins
- *k*: Represents the amount of coin your friend wants
- *m*: Represents the integer value

Input format for custom testing

Note: Use this input format if you are testing against custom input or writing code in a language where we don't provide boilerplate code.

- The first line contains *3* space-separated integers *N, K, and M.*

Output format

Print the answer.

Constraints

$1 \leq N \leq 10^3$
$1 \leq K \leq 10^2$
$1 \leq M \leq 10^3$

Sample input

```
4 2 2
```

Sample output

```
2
```

Explanation

Given

- $N = 4$
- $K = 2$
- $M = 2$

Approach

- There are *2* ways :
- 1^{st} Set : {1, 3} with sum 1+3=4 having 4%2=0
- 2^{nd} Set {2, 4} with sum 2+4=6 having 6%2=0